CS533: ML for EDA

CS533: Course information

TAs:
1. Akash Dutta (PhD)
2. Innocent D. Machahari (PhD)
3. Naman Baveja (MTech)
4. Tellabiyyam Abhilash (MTech)
5. Vaibhabi H. Sidam (Mtech)

Grading:
1. Class test – 1 (before midsem) -- 5
2. Midsem -- 20
3. Class test – 2 (after midsem) -- 10
4. Project -- 25
5. Endsem – 35
6. Attendance --
   1. >= 75% -- 5
   2. < 75 % -- FA

Introduction to the semiconductor industry

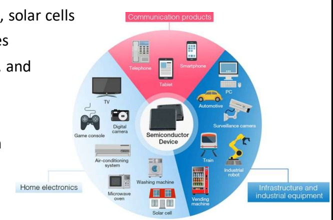
Acknowledgment: Dr. Tahir Ghani (Intel)

Semiconductors

Primary driver of technological innovation

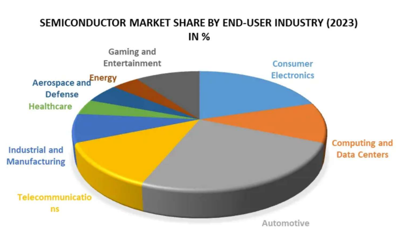
Applications
- Computing – microprocessors, memory, GPU
- Communications – cell phones, satellite systems, and other communication devices
- Energy – voltage regulator, power supplies, solar cells
- Automotive – electric/autonomous vehicles
- Healthcare – Medical imaging, monitoring, and diagnostic equipment

Semiconductor chips are driving AI revolution



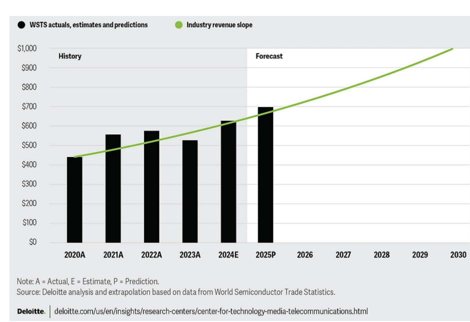
Global demand of semiconductor industry

~75% of the global demand of semiconductors is driven by
- Telecommunication
- PC/Compute
- Automotive
- Consumer electronics



Source: https://www.maximizemarketresearch.com/market-report/semiconductor-market/215503/

Financial growth of semiconductor industry



Semiconductor manufacturing

Integrated device manufacturing model (IDM)

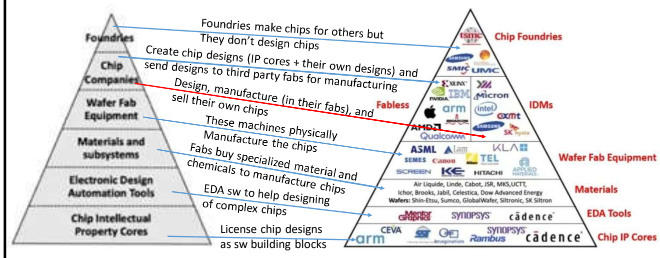
IDMs design, manufacture, and sell their own chips.
This was the traditional model of chip development (Intel, Samsung, …)

Foundry model

Foundry model outsources different stages of production to specialized companies.

- **Fabless semiconductor companies** create the design and products (AMD, Nvidia, Qualcomm)
- **Foundries** are contracted to manufacture the designed chips (TSMC, Samsung, Intel)
- **OSAT (outsourced semiconductor assembly and test)** companies assemble, package, and test the manufactured chips (ASE Holdings, Amkor)

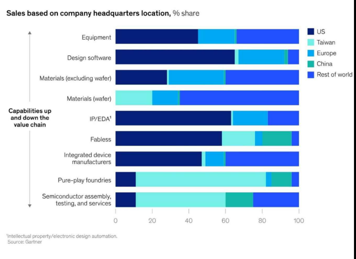
Semiconductor industry segments



Design and manufacturing expertise

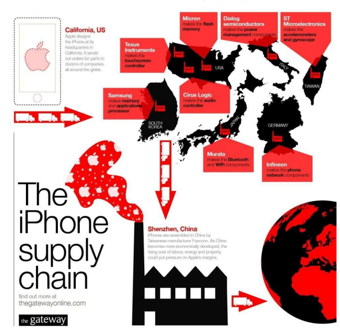
Expertise by location (headquarter)

- Intel (USA) => Desktop & laptop CPUs
- Qualcomm (USA) => Smartphone SoC
- TSMC (Taiwan) => Chips at leading edge Assembly & Test
- ASML (Dutch) => Advanced lithography Equipment
- Samsung (S. Korea) => Memory
- NVIDIA (USA) => Graphics and now AI
- Japan => Specialty chemicals for semi. manufacturing
- Japan & S. Korea => Starting wafers

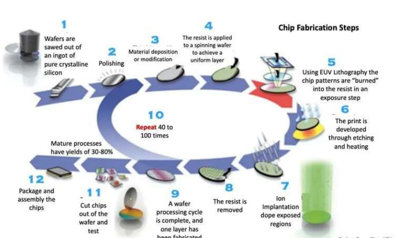


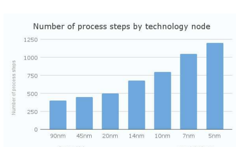
No region has end-to-end capabilities for semiconductor design and manufacturing

The iPhone supply chain



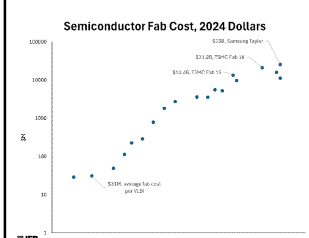
How to make a chip (simplified)





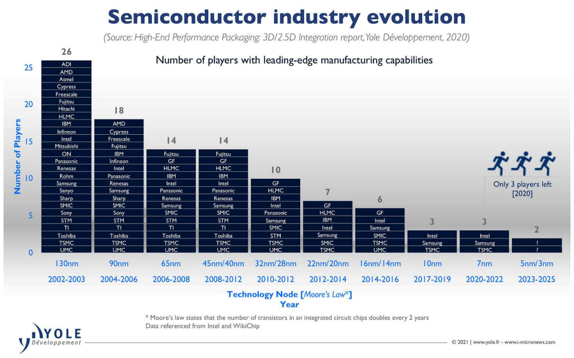
Source: https://kaizoft.com/semiconductor-pricing-a-complex-art-cost-drivers-and-trends/

Semiconductor fab



Intel fab under construction in Ireland

Source: https://www.construction-physics.com/p/how-to-build-a-20-billion-semiconductor



Evolution of the semiconductor industry
From vacuum tubes to AI chips

Acknowledgment: Dr. Tahir Ghani (Intel)

Electronics before 1950

Vacuum tubes used for radio, TV, radar, and computer systems

Issues: Fragile, bulky, expensive, and power hungry


Operating tubes in an audio power amplifier

ENIAC (1946) – first vacuum tube computer
- 18000 vacuum tubes
- 70000 resistors
- 10000 capacitors
- 6000 manual switches
- ~5 million hand-soldered joints
- 18000 square feet
- Power: 150kW

- Capability: add 5000 numbers/second

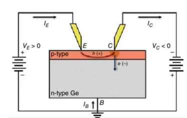

Source:https://cse.engin.umich.edu/about/history/eniac-display/

Discovery of the transistor (1947-48)

Point contact transistor (1947)

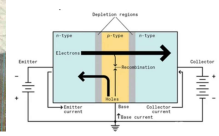

John Bardeen


Walter Brattain



Bipolar junction transistor (1948)


William Shockley



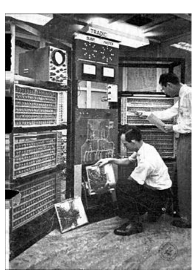
Shockley, Bardeen, and Brattain were awarded the Nobel Prize in physics (1956)

The first computer built using transistors (1954)

During 1950, semiconductor devices gradually replaced vacuum tubes in digital computers

Transistors were 20x more expensive than vacuum tubes, but they were smaller, faster, and used nominal power compared to vacuum tubes.

Bell Laboratory (1954): built first computer without vacuum tubes, named **TRADIC**
- 700 point contact transistor
- 10000 diodes
- 1 MHz
- < 100 Watts of power (150kW for ENIAC)
- 3 cubic feet volume vs 1000 square feet for ENIAC

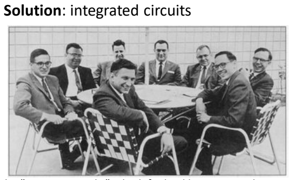

Source: en.wikipedia.org/wiki/TRADIC

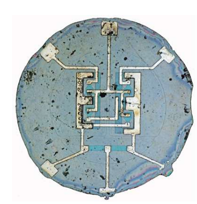
Transistor computers and integrated circuits

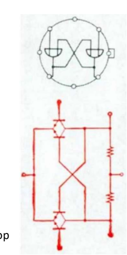
In 1950, circuits were built by connecting (soldering) discrete components, such as transistors, resistors, and capacitors

Issues: Interconnecting thousands of components was tedious and error-prone

Solution: integrated circuits

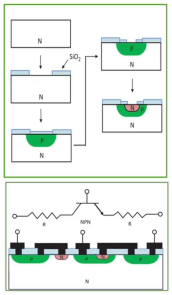

The "Traitorous Eight" who left Shockley Semiconductor Laboratories to form Fairchild Semiconductor Corp. in 1957 (clockwise around table from far left): Jean Hoerni, Julius Blank, Victor Grinich, Eugene Kleiner, Gordon Moore, C. Sheldon Roberts, Jay Last, and (in front) Robert Noyce.

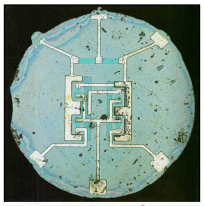


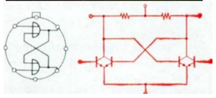

First planar IC: Fairchild's 1960 flip-flop using Jean Hoerni's planar process

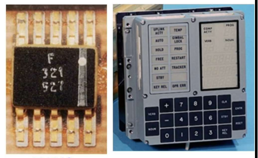
Integrated circuits and Apollo program

Monolithic integrated circuit (1959) | Fairchild uLogic RTL IC (SSI) (1959) | Apollo auto guidance computer (AGC) (1966-1970)

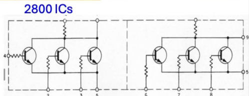








2800 ICs

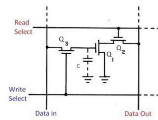


AGC dual 3-input NOR gate schematic (Bipolar Transistors) – Resistor Transistor Logic (RTL)

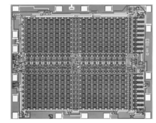
Two key milestones: First DRAM and Microprocessor

Intel 1103 DRAM (1973) – First DRAM

DRAM memory cell (i1103)



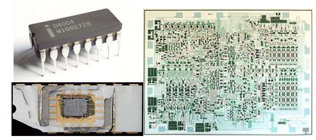
1024 bit DRAM



With the 1103, Intel established semiconductor memory as the new standard technology for computer memory (replacing magnetic core memories)

The rest is history!

Intel 4004 – First ever CPU on a chip



With the 4004 (2,300 transistors), Intel demonstrated the first CPU with all key compute units implemented on a single chip, serving as a general-purpose processor that could execute a range of functions by modifying a software program.

Source: https://en.wikipedia.org/wiki/Intel_4004

Moore's Law (1965)

Gordon Moore in 1965

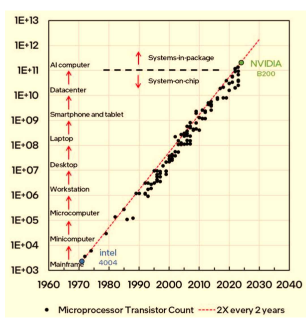
The number of transistors on a microchip roughly doubles every two years.

NVDIA B200 – 2024 -- ~200B transistors

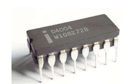
27 years $2^{27} \times 2000 \sim 250B$

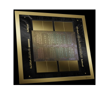
Intel 4004 – 1971 -- ~2000 transistors

Exponential growth of ~2x every 2 years for over 60 years



50+ years of human ingenuity at best





Specification	Intel 4004 (1971)	NVDIA B200 (2024)
Transistor count	2300	208 B
Process node	10000 nm	4nm
Chip dimensions	12 mm^2	~1600mm^2
Tightest pitch	10000 nm	28 nm
Top frequency	740 kHz	~1.8GHz
Total power	0.45 W	1000 W
TOPS	0.000092	20000